

Studies on dry root rot of Chickpea caused by *Rhizoctonia bataticola* (Taub.) Butler

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In

AGRICULTURE

PLANT PATHOLOGY

By

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2018

CERTIFICATE – I

*This is to certify that the thesis entitled “**Studies on dry root rot of Chickpea caused by Rhizoctonia bataticola (Taub.) Butler** ” submitted in partial fulfillment of the requirements for the Degree of **MASTER OF SCIENCE** in **Department of Plant Pathology** of **Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalaya, Gwalior**, is a record of the bona-fide research work carried out by **Ms. Rekha Shrivastav** I.D. No. RA/GW/197/2012 under my guidance and supervision. The subject of the thesis has been approved by the Student’s Advisory Committee and the Director of Instructions.*

No part of the thesis has been submitted for any other degree or diploma or has been published. All the assistance and help received during the course of this investigation has been acknowledged by the scholar.

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CERTIFICATE - II

*This is to certify that thesis the entitled “**Studies on dry root rot of Chickpea caused by Rhizoctonia bataticola (Taub.) Butler**” submitted by Ms. Rekha Shrivastav I.D. No. RA/GW/197/2012 to the Rajmata Vijayaraje Scindia Krishi Vishwa Vidyalyaya, Gwalior in partial fulfillment of the requirements for the degree of Master of Science in **AGRICULTURE** in the department of **Plant Pathology** has been accepted after evaluation by the External Examiner and approved by the Student’s Advisory Committee after an oral examination of the same.*

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Chapter - I

INTRODUCTION

Pulses on account of their vital role in food security and soil ameliorating properties have been an integral part of sustainable agriculture since ages. Pulses are important source of protein, high in fiber content and provide ample quantity of vitamins and minerals. Pulses are important component to sustain the agriculture production as the pulse crops possess wide adaptability to fit into various cropping systems, improve the soil fertility being leguminous in nature and physical health of soil while making soil more porous due to tap root system.

In India, Chickpea is grown almost in all parts of the country mainly as a rainfed crop (68% area). During 2017-18, Chickpea production has been estimated to be about 11.16 million tons, which is about 45% of the total pulse production (24.5 m t) in India. Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Andhra Pradesh, Karnataka, Chhattisgarh, Bihar and Jharkhand contribute more than 95% to the total chickpea production in the country. In India, both desi and kabuli type chickpea varieties are grown.

There has been an impressive growth in area, production and productivity of chickpea in India during the past decade. During 2017-18, chickpea production is expected about 11.16 million tonnes attaining highest peak in production in the history of its cultivation in India. Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Andhra Pradesh, Karnataka, Chhattisgarh, Bihar and Jharkhand are major chickpea producing states contributing more than 95% to the total chickpea production in India. Madhya Pradesh is the single largest producer in the country, accounting for over 38% of total production.

India is the largest producer of chickpea with a share of about 66% in area and 65% in the production of chickpea in the world. India continued to remain a major importer of desi chickpea, but as emerged as a major exporter of Kabuli chickpea during the past decade.

The major chickpea growing countries are India, Pakistan and Turkey in Asia, Ethiopia in Africa, California and Washington State in the USA, Mexico and Australia. It is cultivated in about 8.35 million ha with an average annual production of 7.17

million tones and 589 kg/ha productivity (Anon, 2015). In Madhya Pradesh, the crop is cultivated in an area about 3.02 million ha with production of 3.27 million tones and 1083 kg/ha productivity contributing about 40 % production in the national basket (Anon., 2015).

Chickpea is attacked by air borne, seed borne, and soil borne pathogens. More than 75 pathogens have been reported to infect chickpea (Nene *et. al.*, 1989). Diseases are the most serious constraints to chickpea productivity causing up to 100% losses. Environmental factors and intensity of abiotic stresses are known to compound the occurrence and severity of diseases. Though many diseases are reported, only a few diseases such as Wilt (*Fusarium oxysporum* f.sp. *ciceris*), Dry root rot (*Rhizoctonia bataticola*), Collar rot (*Sclerotium rolfsii*), Wet root rot (*Rhizoctonia solani*), Ascochyta blight (*Ascochyta rabiae*), Botrytis grey mould (*Botrytis cinerea*) and Chickpea stunt may cause major losses and prevent farmers from realizing the potential yield of chickpea.

The Dry root rot (DRR) of chickpea caused by necrotropic fungus *R.bataticola* (Taub.) Butler is emerging as a serious threat to the chickpea production worldwide (Pande and Sharma, 2010). Dry root rot generally appears during late flowering and podding stages.

The management of the disease can be done through cultural, chemical, biological methods and use of resistant varieties. In the absence of resistance / tolerant variety it is difficult to manage the disease caused by soil borne pathogens because of complex soil environment of physical, chemical and biological origin.

Present investigations were, therefore, undertaken with the following objectives:

1. To evaluate the culture media against growth of *Rhizoctonia bataticola*.
2. Screening of cultivars of chickpea against dry root rot pathogen under sick micro plot.
3. To see the effect of date of sowing on the development of root rot of chickpea.
4. To evaluate *In-vitro* safer chemicals and botanicals against *Rhizoctonia bataticola*.

Chapter - II

REVIEW OF LITERATURE

History and occurrence of *R. bataticola*

R. bataticola (Taub.) Butler as a plant pathogen was recognized by Halsted (1890). The first report of occurrence of dry root rot in chickpea along with wilt was made by Padwick (1948).

Ashby (1927) accepted *Macrophomina* and rejected the binomial *Macrophomina phaseoli* and proposed a new binomial *M. phaseolina* as the pycnidial stage of *R. bataticola* on the basis that *M. phaseoli* was the earliest applicable binomial. He was not aware that Tassi had earlier described *Macrophomina phaseolina*.

Small (1926) recognized the identify of this fungus with the forms from India, Egypt, and America and abandoned the name *Rhizoctonia lamellifera*.

Haigh (1930) suggested that *R. bataticola* be used for *sclerotial* isolate and pycnidial strains should be called *M. phaseolina*. Goidanich (1947) examined the original material of Tassi and compared it with *Macrophomina phaseoli*, *M. corchori*, *M. cajani*, *M. sesami*, *M. philippinensis*, *Dothiorella cajani* and *D. phaseoli*. According to the International Code of Botanical Nomenclature, the binomial *Macrophomia phaseolina* was the valid name for the pycnidial stage of *R. bataticola*. *Macrophomina phaseolina* is a soil born pathogen, first reported on chickpea from India (Mitra, 1931).

The most important taxonomic work on the genus *Rhizoctonia* in the 19th century is that of Tulasne brother's (1862) they published the first extensive morphological study on the genus *Rhizoctonia*.

Butler (1925), however, considered that the fungus should be named, as *Rhizoctonia* and made the combination *Rhizoctonia bataticola* (Taub). Jones (1925) gave the name *Rhizoctonia bataticola* (Taub).

The fungus is mainly a soil dweller and spreads from plant to plant through irrigation water, food and implements and cultural operation. The sclerotia and pycniospore may also become air borne and cause further spread of the pathogen (Rangaswami and Mahadevan., 2008).

Recent reports indicates that dry root rot (*Macrophomina phaseolina*) is emerging a potential threat to chickpea production.(Ghosh *et al.*, 2013)

Culture media

Sharma *et al.* (2013) evaluated five culture media viz., Asthana and Hawkers media, Potato dextrose agar, Czapek's Dox agar, Corn meal agar and Richard' s agar to findout the appropriate growth of *R. solani*. Maximum growth of *R. solani* was found in Potato dextrose agar followed by Czapek's Dox agar, Corn meal agar, Richards agar and Asthana and Hawkers medium.

Rhizoctonia bataticola grew best on Cotton root extract agar, PDA, Richard's agar and Sabouraud's agar. Sclerotial production was excellent on the last 3 media and on Czapek's agar.(Kulkarni *et al.*, 1992)

Zhou *et al.* (2002) evaluated 12 culture media for the mycelial growth and sclerotial number of *R. solani*. Results showed that cooked Rice straw solution agar, water agar, PDA and Maltose agar were better media for *R. solani*. Richard's agar was the best for sclerotial formation, and the highest sclerotial number was obtained in this medium.

Singh and Singh (2007) cultured *R. solani* on B-R-Agar, Brown's Agar, Flentze's soil extract Agar, Potato Dextrose agar (PDA), Water agar, and Potato marmite agar. Maximum fungal colony diameter (89.7 mm) was recorded on Potato dextrose agar, which was at par with PDA + Rice leaf extract agar. The medium supporting best growth were Flentza's soil extract agar, Brown's agar and Potato marmite agar. Water agar supported the minimum growth of *R. solani*. Maximum sclerotial production was observed on PDA, followed by Richard's Agar, Soybean decoction sucrose agar and B-R agar. PDA was the best basal medium in terms of maximum sclerotial production and viability.

Salunkhe *et al.* (2009) carried out an experiment to know the effect of nutritional and physiological factors on growth and sclerotial formation of *R.bataticola* isolates. Out of six media, tested, PDA supported the growth and sclerotial formation of *Rhizoctonia bataticola* followed by Czapek's Dox, Richard's, Oat meal and Peptone agar media.

Screening of chickpea entries

Krishnamohan *et al.* (1981) screened twenty chickpea entries under field conditions and artificially in the green house, using a 1-6 score, BG 205 and BG 206 found moderately susceptible in the green house, and highly resistant under field conditions against *R. bataticola*.

Three chickpea cultivars did not related to their resistance to *R. bataticola*. Carbohydrate content was higher in the susceptible cultivar in comparison to resistance cultivar. Singh *et al.* (1982)

Singh and Mehrotra (1982) reported few lines resistant to *R. bataticola*. Cultivars BG-203, G-543 and Hare chhole exhibited resistance reaction when grown in infested soil.

Reddy *et al.* (1990) tested varieties for their resistance to wilt and different root rots of chickpea and found that the mortality of variety JG-62 was 100% whereas, Avrodhi and ICC-48 showed 20% mortality.

Baker and Ahmed (1991) evaluated the resistance against 90 genotypes of chickpea in field infested with wilt, and dry root rot pathogens and found that ICC 12263 was most resistant.

Jayanti and Bhatt (1993) evaluated 21 chickpea varieties, which were sown in contaminated soil. *R. bataticola* caused seed rot within 24 hours in NEC 874 and EG 234, Bold 2375, BG 209, JG 62, JG 315, ICC 3357 and JG 1133 developed necrotic lesion 3 to 5 cells deep on the hypocotyls region within seven days. Variety RSG 44, AGC 677, NEC 41, GL 269, JG 74, ICC 8983 and ICC 5003 developed only superficial necrosis along the hypocotyle region. BGM 416, BG 416, ICC1376 and ICC 113314 were resistant. Resistant cultivars had a greater number of lateral roots during early growth phases.

Forty-five chickpea cultivars were evaluated for resistance to dry root rot *R. bataticola* (*M. phaseolina*). Prajapati *et al.*, (2003) Among these, the following cultivars were rated as resistant which are H 355, JG 315, Pusa 212, G 543, ICC 32, BG 244, Phule G5, IICC 2644, 10384, 11224, 10630 and 11332, ICCV10, ICC numbers 12441, 12263, ICCL 81010, 81002, Avrodhi, ILC 702 and ILC 200. Five cultivars were moderately resistant (GCP-9504, Phule G 96022, 96020 and GL 91059) and 9 were tolerant (JG 22 and 23, and Phule G -96022, 95123, 93208, 92028, BCP 91 and BCP 28).

Pandey *et al.*, (2004) screened twenty-nine chickpea germplasm accessions, ten cultivars, eight advanced breeding lines for resistance to dry root rot, caused by *R. bataticola*. One germplasm accession (ICC 14395), a cultivar (ICCV 2) and an advanced breeding line (ICCX830203-BH-BH-11H) were resistant to dry root rot. Of the remaining lines, 22 were moderately resistant, 19 susceptible and 3 highly susceptible. The disease severity in the two susceptible lines (BG 212 and ICC 12267) was used as the control and rated 9.

Mishra *et al.* (2005) tested 470 chickpea germplasm lines against *R. bataticola* and found KG 86, KWR 4, KWR 108 and KWR 277 as resistant genotypes.

Chaturvedi and Dua (2009) screened twenty five resistant cultivars. Cultivars KPG 59, Radhey and K 50 were found to be most promising against dry root rot of chickpea.

Twenty three isolates of *R. bataticola* causing dry root rot of chickpea (*C. arietinum*) were collected from 10 different major chickpea growing states of India. Isolates were highly variable in their morphological and cultural characters as well as in their pathogenicity/virulence. (Aghakhani and Dubey, 2009) Pathogens were grouped into 6 pathotypes. The pathotype groups were related to agro ecological region of the country. The most virulent isolate (RBI from Bangalore, Karnataka) was fast growing and produced largest sclerotia. A set of cultivars was proposed for the first time for differentiating the pathotypes of *R. bataticola* causing dry root rot of chickpea.

Gupta *et al.* (2012) assessed one hundred seventy chickpea genotypes for locating new and better sources of resistance against dry root rot through blotter paper technique and in multiple disease sick fields. The susceptible cultivar BG 212 was used which exhibited 100 per cent mortality i.e. rating 9 in 1-9 scale. Out of 170 accessions, sixty eight genotypes exhibited resistant reaction (<10% mortality), out of which twenty eight were the promising lines namely (JG 1-14, 2-125, 2-4-110, 14-11, 14-10, 2001-13, 2001-13, 2001-18, 2001-80, 2001-115, 2002-20, 2003-95, 2003-14-16, 2004-110, 210, 9605, 1-9, 99-115, 2001-04, 2003-14-2, JG 2000-07, JSC 37, MPJG 89- 11551, MPJG 89-9023, CSJ 592 and Rajas). Resistant genotypes had

white healthy root system with greater number of lateral roots. All other lines having >20% mortality showed necrotic root lesions leading to extensive root rotting.

Iftikhar and illyas (2000) evaluated 108 chickpea germplasm cultivars/lines for sources of resistance against *M. phaseolina* infection and disease assessment on the basis of external expression of dry root rot symptoms revealed that none of the tested lines was highly resistant. Only one line (97112) was resistant and another line (97104) was moderately resistant. Six lines viz., 97107, 97113, 97119, 97154, 97156 and 97157 exhibited moderately susceptible reaction, while 18 and 81 lines were susceptible and highly susceptible, respectively. When the screened lines were uprooted from pot soil, 150 days after sowing, for observation on root lesions, sclerotial plugging in xylem vessels of roots and collar region, none of the tested lines could be rated as highly resistant, resistant or moderately susceptible. Only three lines (97104, 97112 and 97144) were rated as susceptible. All the remaining lines were graded as highly susceptible.

Vijay Mohan *et al.* (2006) evaluated twelve chickpea cultivars for resistance against dry root rot disease (*M. phaseolina*) under natural and artificial conditions. Under natural conditions, the mean disease incidence was lowest (8.0%) in G-543, followed by BG360 and GNG-1365 which recorded mean disease incidence of 9.6 and 9.8%, respectively.

Jayalakshmi *et al.* (2009) were conducted a pot experiment for screening of chickpea breeder promising entries (12) and susceptible check JG- 62 during rabi season of 2004- 2005 and 2005- 2006. Out of 12 genotypes tested against root rot disease 4 were found resistant, 2 were moderately susceptible, 3 were susceptible and 3 were highly susceptible to dry root rot disease.

Sixty germplasm lines of chickpea were screened for their resistance against dry root rot disease in pot in Jammu Kashmir. Only nine lines namely KGD-1189, KGD-1201, KGD-1209, KGD-1215, KGD-1217, KGD-1220, KGD-1221, KGD-1248 and KGD-1289 were found resistant. (Khan *et al.* (2013)

Date of Sowing

Studies on the interaction of date of sowing with varieties and bio agent on dry root rot of chickpea showed that late sowing i.e. 15th Nov. significantly reduced the incidence of dry root rot in all three chickpea cultivars (Prajapati *et al.*, 2003).

Prasad and Ram (2005) studied the effect of weather condition on dry root rot disease in chickpea cv. Pant G 114 and found the lowest disease incidence (18.1 and 17.3%) of dry root rot and the maximum yield (14.8q/ha and 14.9q/ha). Disease incidence was significantly and positively correlated with maximum, minimum and mean temperature and the minimum and mean relative humidity. The maximum relative humidity was non significantly and negatively correlated with disease incidence. However, rainfall was non significantly and positively correlated with disease incidence. Multiple correlation coefficients between disease incidence and weather variable exhibited strong relationship among the different component and at epidemic during period 2001-2003. Result showed that late sowing significantly reduced the incidence of dry root rot in all three chickpea cultivar (17.1%) in JG-9.

Khan *et al.* (2013) reported that the variation in date of sowing was tested as an effective and economic strategy against dry root rot caused by (*R. bataticola*) and wilt (*Fusarium oxysporum f.sp. ciceris*) disease of chickpea.

Under emerging climate change, high temperatures and the resultant rapid loss of soil moisture at the reproductive stage of growth predispose chickpea to outbreaks of dry root rot. Sharma and Pande (2010)

Sharma *et al.* (2013) reported that erratic rainfalls and rise in temperature have become more frequent under the changing scenario of climate particularly in semiarid tropics. Dry root rot (DRR) caused by *Rhizoctonia bataticola* was found as a potentially emerging constraint to chickpea production than wilt. Increasing incidence of DRR indicate strong influence of climate change variables such as temperature and moisture on the development of disease. The DRR incidence was significantly affected by high temperature and soil moisture deficit. Out of five temperature regimes (15^o C, 20^o C, 25^o C and 30^o C) and four moisture levels (40, 60, 80 and 100%), a combination of high temperature (30^oC) and soil moisture content (60%) predisposes chickpea to DRR. The study clearly demonstrates that high temperature coupled with soil moisture deficit is the climate change variables predisposing chickpea to *R. bataticola* infection, colonization and development.

Botanicals and Fungicides

Datar (1999) studied the effect of Garlic (*Allium sativum*) extract on charcoal rot (*M. phaseolina*) in sorghum. Phyto-extracts of eleven plant species against *M.*

phaseolina of green gram was studied by Tandel *et al.* (2010) and revealed that the onion bulb extract produced maximum inhibition followed by extract of Acacia, Ginger, Neem, Garlic and Karanj.

Oluma *et al.* (2002) screened eight plants for their fungitoxicity against *M.phaseolina* using Czapeck-Dox-agar plant extract media. The plants were *Prosopis africana*, *Caesalpin apulcherrima*, *Calotropis procera*, *Nauclea latifolia*, *Eucalyptus camladulensis*, *Chromolaena odorata*, *Azadirachta indica* and *Thevetia peruviana*. Out of these, the extracts of pulverized bark of *P. africana* and leaves of *N. latifolia* inhibited both radial mycelial growth and sclerotial formation by 100%. There was no inhibition of radial mycelial growth with extracts of *C.procera*, *A. indica* and *T. peruviana*.

Jha and Sharma (2008) screened an aqueous autoclaved leaf extract of eighty three plant species *in-vitro* for their fungicidal activity against isolates of *R.bataticola* by Paper disc diffusion assay technique. The study revealed that only a few plant extract viz., *Rannunculus scleratus*, *Xanthium stramonium*, *Ipomoea carnea*, *Ocimum basilicum* and *Eclipta alba* showed varied antifungal activities.

Mandhare and Suryawanshi (2008) reported that *Allium sativum* (15%) inhibited the growth of *F. oxysporum f. sp. ciceri* and *S. rolfsii* completely and of *R. bataticola* by 88.8%. The extract of *Azadirachta indica* inhibited the growth of *F. oxysporum f. sp. ciceri*, *S. rolfsii* and *R. bataticola* by 55.5, 33.3 and 58.8% respectively.

Ammajamma *et al.* (2009) evaluated botanicals and reported that eupatorium was highly effective at 10 per cent (81%) followed by Garlic and Neem leaf extract (50 and 16.67% respectively).

Mandhare and Suryawanshi (2009) screened the antifungal properties of the extracts (at 10% each) of *Azadirachta indica* (leaves), *Ocimum sanctum* [*O.tenuiflorum*] (leaves), *Eucalyptus sp.* (leaves), *Nerium indicum* (leaves), *Allium sativum* (cloves) and *Zingiber officinale* (rhizomes) against *R.bataticola* by Poisoned food technique. The growth of *R. bataticola* was inhibited by 77.77 and 64.44% by the extracts of *Allium sativum* and *A.indica*, respectively. The other extracts were not effective.

Dhingani and Badani (2013) evaluated plant extract against *Macro- phomina phaseolina* causing dry root rot of chickpea. The extract of garlic clove (*Allium sativum* L.) was proved excellent with maximum mycelial growth inhibition (73.0%) and sclerotial formation followed by rhizome extract of turmeric (*Curcuma longa*) (63.98%).

The onion extract was found most effective in inhibiting (100%) the growth of *R. solani* in soybean whereas, Eucalyptus exhibited minimum growth inhibition (3.2%) at 20 per cent concentration. (Nagiand Badani., 2014)

Sangeetha and Shamarao (2013) in the *in vitro* study found that *Trichoderma viride*, *Trichoderma harzianum*, *Azadirachtin* and Neem oil were more effective in inhibiting the mycelial growth of the *R. bataticola*.

Meena *et al.* (2014) explored the efficacy of eight different plant species, *Zingiber officinale*, *Aloe indica*, *Lawsonia inermis*, *Chenopodium album*, *Allium cepa*, *Piper betel*, *Murraya koenigii*, *Parthenium hysterophorus* aqueous extracts against *M.Phaseolina* causing stem rot of jute by using of poisoned food technique. The rhizome extract of *Z. officinale* produced maximum growth inhibition (74.59%) of the pathogen followed by leaf extracts of *A. indica* (63.57%), and leaf extract of *L. inermis* (60.17%) at the 10% concentration.

The phytoextracts of nine plant species viz. garlic, onion, lantana, tulsi, jatropa, ardusi, neem, turmeric and ginger were evaluated *in vitro* by poisoned food technique against *M. phaseolina*. (Savaliya *et al.* 2015). The extract of garlic cloves was proved excellent with maximum (77.65%) inhibiting mycelial growth and scanty sclerotial formation followed by onion bulb extract (63.98 %). While least growth inhibition (32.34 %) was recorded in ginger rhizome extract.

Goyal and Mehrotra (1981) tested 9 fungicides against *R. bataticola in vitro* and on (chickpea) Benlate, Bavistin. Thiophanate metyle and Dyrene were promising. In vitra, Venomyle and Bavistin inhibited fungal growth most.

Sarwar and Raju (1985) tested 8 fungicides *in vitro* an antibiotic Topsin M-70 was highly effective at 1000, p.pm. against *R. bataticola*.

Taya *et al.* (1990) in green house tests, in different ways and found that the best control of *R.bataticola* was given by carbendazin alone or in combinaltion with

Thiram as seed treatment, pre- sowing, soil-drench and seed treatment +drenching after sowing treated seed of chickpea

Peshney *et al.* (1992) reported that thiram, Campton, PCNB moncozeb Iprodione, carboxine, carbendazim Thiobedazole (all 0.2%) and tridemorph (0.07%) effectively controlled growth on Sclerotial germination of *R.bataticola* in vitro. Carbendazine and thiobendazole were most effective and inhibited growth at the concentration of 0.006% In successive cultures, the fungus should increase tolerance to Mancozeb and PCNB and to a lesser extent to Thiram, Campton, carboxin and iprodione. There was no reduction in sensitivity towards carbendazim or thiobendazole.

Malathi and Doraisamy (2003) tested compatibility of fungicides and biocontrol agent viz., Captan, Thiram, carbendazim and *Trichoderma harzianum* individually and in combination against *M.phaseolina*, *In vitro*. It was found that *M. phaseolina* was completely inhibited by Carbendazim and Thiram.

Choudhary *et al.* (2004) observed the effectiveness of four fungicides, i.e. Bavistin (Carbendazim), Antracol (Propineb), Indofil M-45 (Mancozeb + Thiophanate-methyl) and Ridomil MZ (Mancozeb + Metalaxyl), applied at 300, 400, 500 and 1000 ppm, in inhibiting the mycelial growth of *M.phaseolina*, the causal agent of stem and root rot of sesame, *in- vitro* using the poisoned food technique. All fungicides inhibited mycelial growth compared with the untreated control, with Bavistin being the most effective.

Khan *et al.* (2012) a eight fungicides under evaluated laboratory condition against dry root rot of chickpea caused by *R.bataticola* and found that Indofil M-45, Bavistin, Companion, Copper Oxycloiride and Benelate were completely inhibited the growth of the fungus on potato dextrose agar medium as compared to check.

Kumari *et al.* (2012) tested seven fungicides *in vitro* by poisoned food technique and in pots (*in vivo*) against root rot of mungbean and found that Bavistin was the most effective to inhibit mycelial growth of pathogen as well as reducing root rot incidence, followed by Captan, Thiram, Indofil M-45, Vitavax and Raxil, while Copper sulphate was the least effective treatment in both conditions.

Nagamma *et al.* (2012) tested fungicides in vitro against which cause dry root rot of chickpea caused by *Macrophomina phaseolina* Among the systemic

fungicides, cent percent inhibition was recorded in Hexaconazole, Carbendazin, Difenconazole and Propiconazole at 50 ppm.

Jyothi and Muhammad (2016) found that seed treatment with Tebuconazole (2g kg⁻¹ of seed) recorded least disease incidence (8.70 %) with highest yield followed by seed treatment with Tebuconazole (2g kg⁻¹) + Carbendazim (1g kg⁻¹) recorded 15.66 % diseases incidence as compared to control.

Chapter – III

MATERIAL AND METHODS

The investigation has been carried out in the Department of Plant Pathology R.V.S.K.V.V. College of Agriculture, Gwalior (M.P.) during Rabi 2017. The present studies on dry root rot of chickpea were focused on screening of material/cultivars and management aspects. The material used and methods followed are described below:-

Climate

The climate of Gwalior is subtropical. The rainy season normally starts from middle of June after commencement of south-west monsoon and last up to September. Maximum precipitation of rains occurred in the month of July and August. Winter season runs from November to Mid-February and hot summer season from April to Mid-June. October is the transitory month between rainy and winter season.

Soil

The soil of experimental site was alluvial clay loam texture. The available nitrogen, phosphorus and potassium content of soil was 200.7 kg, 10.1 kg and 260.9 kg per hectare, respectively low, medium low to medium, respectively. The water holding capacity of soils is also medium.

Experimental site

The present studies were conducted at field and laboratory of Department of Plant Pathology, R.V.S.K.V.V. ,College of Agriculture, Gwalior during Rabi 2017. Gwalior is situated in Northern part of Madhya Pradesh at an elevation of 211.52 meters from mean sea level and lies between latitude and longitude of 26°14' North and 78°15' East, respectively.

Materials

3.1 Equipments and apparatus

The equipments and apparatus which have been used in the study are given below:- Laminar air flow, BOD incubator, Refrigerator, Autoclave, Glasswares,

Microscope, Hot air oven, pH meter, Electronic balance, Forceps, Inoculation Needle, Cork borer, Blade, etc.

3.2 Cleaning and sterilization of equipments

Corning make glassware were used during the period of investigation. All the glassware were cleaned with detergent powder and then rinsing tap water before use. The sterilization of media was done at 15lbs, pressure for 20 min. Petriplates were sterilized in hot air sterilizer at 180°C for 2 hrs. The isolation chamber was sterilized by alcohol, followed by ultraviolet exposure for 20 min. The other materials used in isolation chamber like forceps, inoculation needle, cork borer, blade, etc. were sterilized by dipping them in alcohol, followed by heating on flame.

3.2.1 Sterilization procedure

A. Sterilization of glass wares

Glass wares were washed in liquid detergent under running tap water, and rinsed with distilled water 2-3 times. These were air-dried and then kept in oven for sterilization at 180°C for 2 hrs.

B. Sterilization of inoculating needle, forcep, cork borer and working table

Clean inoculation needle was sterilized by dipping the loop in spirit and heating over the flame until red hot. The process was repeated 2 – 3 times. Forceps and cork borer were also sterilized in the way of needle. The working table of laminar air flow was disinfected by sweeping with cotton soaked in absolute alcohol and exposing it to UV light for 20 minutes.

C. Sterilization of media and distilled water

Sterilized glasswares were used for dispensing media and distilled water. All media were autoclaved at 121°C, 15 psi pressure for 20 min.

D. Sterilization of laminar air flow

Prior to the day of inoculation, the laminar air flow was saturated with alcohol vapours. At the time of inoculation the laminar air flow chamber was wiped with 70% alcohol or general spirit. Then only required instruments were kept in the chamber and exposed to UV rays for 20 minutes. All the operations viz., transfer, inoculation etc. were done over a gas burner flame.

E. Culture media

All the solid media were sterilized in an autoclave at 15 lbs pressure (p.s.i) for 20 min. Liquid media was sterilized at 15 lbs p.s.i. for 20 min.

3.3 Isolation of pathogen

3.3.1 Preparation of culture medium

For isolation of target pathogen potato dextrose agar (PDA) medium was used. For preparation of PDA, 250 g peeled potatoes were cut into slices and boiled in 500ml of distilled water in conical flask. The extract was strained through a piece of muslin cloth and Twenty g dextrose was added in it. Twenty g agar – agar was melted in 500 ml of distilled water separately and was mixed in potato dextrose solution and the volume was made up to 1000 ml by adding distilled water. PDA was poured in flasks, plugged with non– absorbent cotton plugs and sterilized in an autoclave.

3.3.2 Isolation and purification of the pathogen

Small pieces of infected root tissues were cut with blade, washed well in distilled water to remove dust adhered to the infected pieces. Pieces were dipped in 0.1 per cent mercuric chloride solution for 30 seconds and finally washed well in three changes of sterilized distilled water and dried between two sterilized blotting paper. The bits were then transferred to PDA slants with the help of inoculation needle under aseptic condition and incubated at $28\pm 2^{\circ}\text{C}$. After 72 hrs, fragments of hyphal growth from the growing tips were transferred to fresh PDA slants. Pure culture was made, following repeated hyphal tip transfer. Pure culture was maintained on PDA slants by sub culturing it at 30 days intervals. For preservation of cultures the plugged end of the culture tubes were dipped in melted wax and stored in a refrigerator at $5\pm 1^{\circ}\text{C}$. (Rangaswamy and Mahadevan, 1999)

3.4.1 Methods of inoculation

For inoculating solid media in Petri- plates, 5 mm disc of fungal mycelium 72 hours old culture grown on potato dextrose agar medium was used. It was placed at the centre of the plate in an inverted position, so that it came in direct contact with the surface of the medium. For inoculating liquid media in 100 ml Erlenmeyer flasks

containing 25 ml broth medium, one disc of 5 mm diameter of fungal mycelium was allowed to float on the medium.

3.4.2 Incubation

The inoculated Petri plates were incubated at $28\pm 2^{\circ}\text{C}$ in B.O.D. incubator for required period.

3.4.3 Measurement of radial growth of colony

Radial growth of the isodiametric colonies were measured in two directions at right angles with help of a linear scale. In case of irregular colonies, measurements were recorded at the broadest and narrowest diameter and the average of two different directions were taken growth. In all the cases radial growth was recorded after 168 hrs of incubation. In case of poisoned food techniques, it was recorded after 120 and 168 hrs of incubation.

3.5 Effective medium for the growth of *Rhizoctonia bataticola* under *in-vitro* condition

Seven culture media viz., Water Agar, Czapek's Dox agar, Chickpea leaf potato dextrose agar, Sorghum grain agar, Chickpea potato dextrose agar and Chickpea grain dextrose agar evaluated against *R. bataticola* and compared with check ie. Potato dextrose agar to select out the suitable medium for the growth of fungus. All medium inoculated with 5 mm. disc of 72 hours old culture of *R. bataticola*. After inoculation the plates were incubated at $28\pm 2^{\circ}\text{C}$. The fungal growth of *R. bataticola* on inoculated medium was measured 7 days after inoculation.

Table 3.1: The culture media utilized for testing the efficacy of the growth of Root rot pathogen

S.No.	Name of the medium	Ingredients
1	Potato Dextrose Agar (PDA)	Peeled & sliced potato - 200 g ,Dextrose - 20 g, Agar-agar - 20 g ,Distilled water1000 ml
2	Water agar	Agar-agar - 20 g ,Distilled water1000 ml
3	Czapek's Dox Agar (CDA)	Sodium nitrate - 2 g ,Di potassium hydrogen phosphate - 1 g, Magnesium sulphate - 0.5 g, Potassium chloride - 0.5 g, Ferrous sulphate - 0.01 g , Sucrose - 30 g, Agar-agar - 20 g
4	Chickpea leaf potato dextrose agar	Chickpea leaf – 200 g, Peeled & sliced potato - 200 g, Agar-agar 20 gm, Dextrose 20 gm , Distilled water 1000 ml
5	Sorghum grain agar	Sorghum grain – 200 g, Agar-agar 20 gm, Distilled water 1000 ml
6	Chickpea Potato dextrose agar	Chickpea seed – 200 g, Peeled & sliced potato - 200 g, Agar-agar 20 gm, Dextrose – 20 g, Distilled water 1000 ml
7	Chickpea grain dextrose agar	Chickpea seed – 200 g, Agar-agar 20 gm, Distilled water 1000 ml

In the above mentioned media the ingredients were mixed in the required quantity of water, boiled and there after media was prepare.The supernatant was filtered. The supernatant and melted agar agar was then mixed measured and the volume was restored to 1000 ml with hot water. It was again boiled for five minutes and then dextrose was added. The medium was poured into the flask and culture tubes. The flasks and culture tubes were plugged with non-absorbent cotton plug and sterilized in an autoclave. After this the medium was used under aseptic condition for bioassay of the test fungus.

3.6 Screening of cultivars of chickpea against dry root rot in root rot sick microplots.

For the preparation of liquid solution of *R.bataticola*, Potato dextrose broth was inoculated with 5 mm. mycelial disc of 3 days old culture of *R.bataticola*. Fifteen days after inoculation the broth was filtered through wattman filter paper. The mycelial growth was mixed with water in blender to form solution. The seeds of individual cultivars were taken in small polythene bags placed in a tray and 30 ml of the culture solution was poured in each seed bag. After 20 hrs. the seeds were sown in root rot sick microplots.

Two hundred and sixty lines of chickpea procured from AICRP, Chickpea , Kanpur,U.P. were planted in 4 m single row length in sick microplots . The dry root rot incidence was recorded after flowering and before pod formation stage. The details of the experiment was as follow:-

Entries	-	261
Design	-	RBD
Replications	-	2
Plot size	-	4 m single row
Distance	-	Row to Row 30- cm
Plant to Plant	-	10 cm

$$\text{Root rot incidence} = \frac{\text{Infected plant}}{\text{Total number of Plants}} \times 100$$

3.7 Effect of date of sowing on the development of Chickpea root rot in popular cultivars as influenced by different date of sowing.

A Field experiment was conducted at the field of Department of Plant Pathology, R.V.S.K.V.V., College of Agriculture, Gwalior, during *Rabi* season 2016-17. Gwalior is situated in Northern part of Madhya Pradesh at an elevation of 211.52 meters from mean sea level and lies between latitude and longitude 26°14' North and 78°15' East, respectively. The soil of experimental site was alluvial clay loam texture. The available nitrogen, phosphorus and potassium content of soil was 200.7 kg, 10.1 kg and 260.9 kg low, medium low to medium, respectively. The water holding capacity of soils is also medium.

The experiment consists of three treatment combinations with three date of sowing as main plot viz., D1–15 November 2017, D2-30th November 2017, D3 – 15 December 2017 and three varieties viz., V1- JG 16, V2-JAKI 9218 and V3 -JG130 as sub-plots. The design of the experiment was split plot with three replications. The plot size was 5x2.4 m and total number of plots were 27. Sowing was done in furrows at 30 cm between rows and 10 cm between plants with in the row. Basal dose of 20 kg nitrogen through urea 40 kg Di- ammonium phosphate and 20kg Murate of potash was applied uniformly to all plots. The incidence of dry root rot, yield and 100 grain weight were recorded.

3.8 *In-vitro* evaluation of botanicals and chemicals against *Rhizoctonia bataticola*.

3.8.1 Collection of botanicals:

Datura stramonium was collected from College of Agriculture, Gwalior Campus, *Allium sativum* (bulb) and *Zingiber officinale* (rhizome) were purchase from the market.

3.8.2 Preparation of plant extract:

For the preparation of powdered extract, the botanicals leaves/rhizome/clove were thoroughly washed in ordinary tap water and dried in oven at 60⁰ C for two consecutive days. The rhizomes and bulbs were cut into small pieces before drying. After drying the leaves/rhizome/clove were powered through grinder and sieved through fine sieve (52 mesh) and stored in the airtight containers. Requied quantity of the powdered botanical was soaked in double quantity of distilled water in conical flask and plugged with non absorbant cotton for over night. Next day the solution was filtered. The extract was used @ 20 ml, solution of the respective botanical extract was incorporated into 80 ml Potato dextrose agar medium and pinch of Streptomycin sulphate was mixed just before pouring.

Table- 3.2: List of botanicals:

S. No	Botanical name	Plant part used	Percent /100ml	Common name	Family	Chemical constituents
1.	<i>Datura stramonium</i>	Leaves	20	Datura	Solanaceae	Atropin, hyosine, hyoscyamine.
2.	<i>Allium sativum</i>	Clove	20	Garlic	Liliaceae	Allicin, di-allyl disulphide, di-allyl tri-sulfide allyl propyl, di-allyl polysulfide.
3	<i>Zingiber officinale</i>	Rhizome	20	Ginger	Scitamina ceae	Zingiberine, gingerol, camphene, phellandrene, cineol & borneol, gingerin

Three selected and effective botanicals were evaluated in the laboratory against *R. bataticola* by poisoned food technique (Nene and Thapliyal, 1993). The details of the botanicals used in the present study are summarized in table-3.2.

Twenty per cent concentration was used for each botanical. Four replications were maintained for each concentration of the botanicals was added in PDA and sterilized in autoclave. Twenty ml of the medium was poured in each pre-sterilized petri plate. A seven mm disc of seven days old culture of *R. bataticola* was placed in the centre upside down on the medium and the plates were kept in upside down position. The medium without botanical served as control. The radial growth of the colony was measured after seven days growth under different botanicals and control.

3.8.3 *In-vitro* evaluation of fungicides for the management of *R. bataticola*

In laboratory experiment, Seven fungicide (Table-3.3) were evaluated *in vitro* against *Rhizoctonia bataticola* by poisoned food techniques (Nene and Thapliyal, 1979). The Required quantity of the fungicides was added in PDA and sterilized in

autoclave. 20 ml of medium was poured in each pre-sterilized Petri plate. A seven mm mycelium disc of seven days old culture of *Rhizoctonia bataticola* was placed on the poured plate and the plates were kept in upside down position to maintain the growth of pure culture. The medium without fungicides served as control. The radial growth of the colony was measured at started from three days after inoculation

Table-3.3: List of Fungicides

Fungicides were evaluated against *Rhizoctonia bataticola* under *in-vitro* condition by adopting poisoned food technique.

Percent inhibition over control was calculated by following formula suggested by Bliss (1934)

$$\text{Percent inhibition} = \frac{C-T}{C} \times 100$$

where, C = Control, T = Fungal growth

Name of Fungicides tested against *R. batatiicola* (In-vitro)

S.No.	Common Name	Chemical name	Trade Name
1	<i>Carboxin</i>	3-(3-5- dichlorophenyl) –N-(1-methylethyle) -2-4dioxo-1-lemadazolidine carboximide	Vitavax power 75% WP
2	Carbendazim	2-(methoxy- carbomyl benzimidazole	Batistin 50% WP
3	Propiconazole	1-(2,4 di chlorophenyl) -4-ropyl-1,3-dioxolan-2-methyl – H-1,4-triozole	Tilt 25 EC
4	Mencozeb	Manganese zinc ethylene bis dithiocarbomate+zinc	Indofil M 45 WP
5	Hexaconazole	(RS) -2(2,4-dichloro phenyl) -1-(14-1,2,4-trizole-1 yl) hexane-2-1	Contaf 5% EC
6	Copper oxychloride	Copper oxychloride	Blitox 50 WP
7	Mencozeb + Copper oxychloride	Manganese zinc ethylene bis dithiocarbomate + zinc + Copper oxychloride	-

Chapter – IV

RESULTS

The present investigations on various aspects of dry root rot of chickpea caused by *Rhizoctonia bataticola* (Taub.) Butler [Pycnidial stage: *Macrophomina phaseolina* (Tassi) Goid] included, evaluation of culture media against growth of *R. bataticola*, effect of different date of sowing, screening of chickpea cultivar, evaluation of selected botanicals and fungicides against *R. bataticola* in *in-vitro*. The results obtained on these aspects are presented here under.

4.1: Symptomatology

Symptoms of dry root rot are most commonly observed in chickpea during post-flowering stage. The symptoms of the disease were yellowing of the leaves within a day or such leaves drop and plant completely dried within a week after the appearance of the first symptom. The seedlings can also get infected. Drooping of petioles and leaflets is confined to the top of the plant. Sometimes when the rest of the plant is dried, the topmost leaves are chlorotic. The leaves and stems of affected plants are usually straw colored but in some cases the lower leaves and stems are brown. If the plant were pulled out from the soil and examined the basal stem and main root system of diseased plant showed extensive rotting with most of the lateral roots destroyed (Plate-2). The tissues were weakened and break off easily and shredding of bark and lateral root. Dark minute sclerotial bodies can be seen on the wood of the root exposed and inner side of the bark or wherever split open the collar region vertically.

4.2 : Identification of pathogen

On the basis of morphological, and cultural characters the fungus responsible for dry root rot of chickpea was identified as *Rhizoctonia bataticola*.

On PDA at 25°C the culture of *R. bataticola* was black with appressed growth (Plate-13). Hypha measured 14.32 x 6.45 µm and sclerotia were irregular measured 160.35 x 151.87 µm in diameter, black in colour.

4.3 : Evaluation of Culture medium

In the study, seven culture media viz Potato dextrose agar, water agar, Czapek's Dox agar medium, Chickpea leaf potato dextrose agar medium,

Sorghum grain agar medium, Chickpea potato dextrose agar medium and Chickpea grain dextrose agar medium were evaluated to find out the appropriate growth of *R. bataticola*. Mycelial growth was recorded at 7 days after inoculation. The results are presented in Table-4.1 (Plate-3 and Fig-1).

Significantly maximum mycelial growth of *R. bataticola* was observed in Potato dextrose agar (86.66 mm) followed by Chickpea leaf potato dextrose agar (73.33 mm), Chickpea potato dextrose agar (58.33 mm) which were found at par with Czapek's agar medium (54.00 mm). Minimum mycelium growth was observed in Chickpea grain agar medium (32.00 mm) followed by Sorghum grain agar medium (37.66 mm), and Water agar medium (41.66 mm)..

Table 4.1: Evaluation of culture media for the growth of *R. bataticola*

S.No.	Culture media	R1	R2	R3	MEAN
1	Potato dextrose agar	85	85	90	86.66
2	Water agar	45	40	40	41.66
3	Czapek's Dox agar	55	52	55	54.00
4	Chickpea leaf potato dextrose agar	75	80	80	73.33
5	Sorghum grain agar	35	40	38	37.66
6	Chickpea potato dextrose agar	55	60	60	58.33
7	Chickpea grain dextrose agar medium	30	33	33	32.00
	CD				4.51
	Sem $\pm$				1.47

4.4: SCREENING

A total of 261 entries of chickpea were evaluated under natural field condition to find out their reaction against dry root rot of chickpea. The observations were recorded at weekly interval and final disease incidence was recorded before harvesting. It is clear from the recorded data that 4 entries are resistant, 3 entries

are moderately resistant, 50 entries are moderately susceptible, 123 entries are susceptible and remaining 81 entries were highly susceptible. The results are presented in Table-4.2 (Plate-4)

Table-4.2: Reaction of chickpea entries against dry root rot (2017-18)

Reaction	No. of entries	Entries
Resistant	4	JG 41, GNG 2215, BG 3054, GJG1205
Moderately resistant	3	HK 14-25, I201PC 3-74, DIBG 201
Moderately susceptible	50	GJG 1318, GNG 2300, GJ 315, H12-26, GJG 1320, IPC 2011-141, JG 315, GNG 2207, GL 13042, JG 2016-43, PG 214, AKG 1303, RKG13-75, BG 372, Pusa 547, NBeG740, CSJ 824, H 13-12, Phule G 08802, GJG 1421, Vijay, RSG 888, BGD 133, H 12=15, GL 29078, CSJ 882, HK 4, NBeG 440, NBeG 458, NBeG, 790, CSJK 43, Phule G 0914, RVSSG 52, Annegiri, RKD-1, MABC-WR-SA2, DIBG 203, RVSS 51, ICCV 10, RKD 4, NBeG 506
Susceptible	123	H 12-26, BRC 3, GNG 1581, H 12-1, PG 170, RKG 13-380, GNG 2325, PG 177, BG 3076, CSJ 907, Phule G 0819, NBeG 738, GJ 2016-44, DBGV 206, CSJ 866, BG 3075, NBeG 776, GNG 2338, GJG 1416, Phule G 0818, H 13-36, Phule G 13110, H 12-62, NDG 14-24, GNG 2304, BG 3068, GNG 2299, RSG 963, GL 12003, PG 186, RKG 13- 515, IPC 2012-49, IPC 2012-98, GNG 2304, GBeG 778, RKG 13-180, GL 29098, PG

		0104, GNG 2261, Pusa 547, BG 372, IPC 2010-62, BG 372, GNG 2144, PG 178, JG 2016-45, CSJ 887, H 12-29, RSG 957, RG 2011-06, RSG 963, DC 16- 1, GNG 2331, Phule G 0805, H 13-03, PG 160, RVSSG 45, BG 3080, Phule G 13116, NBEeG 778, GJG 1316, IPC 2013-70, GNG 2346, BG 3078, JAKI 9218, RKG 13-393, RSG 931, DBGV 209, JG 11, RKG 13-460, RWK 318, HK 13-309, RVSSG 36, Shubra, Phule G 12313, RKGK 13-271, PG 95311, RKGK 13-261, GNG 2281, RVSSG 30, GLK 28127, BGD 128, GNG 1969, RVSSG 31, KAK 2, BG 3084, RKGK 13-3169, IPCK 2012- 143, RVSSG 46, HK 13-112, RKGK 13-266, Phule G 14308, IPCK 2011-37, JGK 2016- 25, NBeG 458, RVSSG 37, BG 3073, Phule G 0517, NBeG 399, CSJK 139, NBeG 510, GLK 05-43, BG 3085, DBGV 3205, CSJ 303, RKGM 1, PG 159, BG 3082, Phule G 08108, JG 2016-24, BG 3081, RVSSG 48, RKGM 2, HC 5, NBeG 03105, DBGV 210, GL 14107, BG 3062, PG 187, KWR 108, NAeG 786,
Highly susceptible	81	DCP92-3, AKG 1109, GNG 2300, BDNG2015-1, NDG 14-11, PG 172, NBeG807, GCP 101, Phule G 0405, GCP 105, GNG 2264, NDG 15-6, H 12-63, IPC 2013-21, RG 2011-02, RVSSG 42, IPC 2012-108, BRC-1, GL 13001, BGD 138, GNG 2299, H 12-55, NBeG

		507, RVSSG 38, PG 0109, RSG 963, JSC 56, IPC 2013-33, GNG 2327, BG 3079, GL 14012, IPC 2010-217, GL 14002, JG 2016- 9605, ICCV 10, AKG 1216, GNG 2340, BG 3077, CSJ 515, NBeG 779, Phule 0809, IPC 2010-219, NBeG 620, GAG 1423, BG 3073, Vidhan Gram 1, GNG 2285, RVSSG 24, CSJK 114, RVSSG 30, NBeG 529, JGK 2016-13, CSJK 138, GNG 2325, CSJK 126, BG 3083, HK 13-105, BG 3086, DBGV(K) 506, GLK 14311, Phule G 14508, CSJ 515, DBGV 204, RCBM 3, Phule G 0916, CSJ 1005, H 12-17, HK 2, IPCK 04-29, MABC-WR-SA 1, NAeG 1004, DIBG 204,
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4.5: Date of sowing

The progress of root rot of chick pea was measured in three varieties under three different date of sowing. Maximum disease incidence was recorded in JG-16 on 15 December sowing and minimum in JG-130 on 15 November sowing. On 15 November sowing minimum disease incidence was recorded in JG -130 (10.4) followed JAKI -9218(17.1) and JG -16(17.1).

On 30 December sowing, minimum disease incidence was recorded in JAKI-9218(13.9) followed by JG -130(14.1) and JG-16(19.7). These data clearly indicates that early sowing decrease the root rot disease of chickpea in comparsion to late sowing.

The maximum incidence of *R. bataticola* was recorded in JG-16 (25.90 %) and minimum on JAKI 9218 (14.50 %).

The maximum disease incidence (21.2 %) was observed in 15th December sowing and minimum disease incidence (12.87 %) in 15 November (12.87 %) sowing. On the basis of interaction of varieties and dates of sowing variety JG 16 (25.9 %) showed maximum disease incidence in 15th December sowing and minimum in JG-130 (10.4 %) under 15th November sown crop. The effect of cultivars and dates of sowing affected the number of pods/ plant. The maximum was observed with cultivars JAKI-9218 (32.33), JG-130 (26.22) and JG-16 (24.11). Maximum pods /plant (32.22) were observed on 30th November sown crop followed by 15th December sown crop whereas it was minimum (24.11) in 15th November sown crop. Interaction of varieties/cultivars and dates of sowing revealed the maximum pods/plant were observed on cultivar JAKI-9218 under 30th November sown crop whereas it was minimum (18.33) in JG-130 under 15th December sown crop.

Table 4.3: Influence of varieties and date of sowing on occurrence of dry root rot and yield and attributing parameters in chickpea

Varieties	Date of Sowing	Disease incidence (Per cent)	Number of pots/ plants	Yield (Q/ha)	100 grain weight (g)
JG-16	15 Nov.	17.1	21.33	15.27	15.85
	30 Nov.	19.7	26.67	15.00	16.83
	15 Dec.	25.9	24.33	8.41	15.83
Mean		20.90	24.11	12.89	16.17
JAKI-9218	15 Nov.	11.1	26.33	19.16	26.63
	30 Nov.	13.9	39.00	15.55	26.13
	15 Dec.	18.5	31.67	10.61	21.90
Mean		14.50	32.33	15.11	24.89
JG-130	15 Nov.	10.4	18.33	12.77	29.87
	30 Nov.	14.1	31.00	12.22	26.93
	15 Dec.	19.2	29.33	10.48	24.60
Mean		14.57	26.22	11.82	27.13
For interaction on between VXD					
CD(P=0.05)	1.16*		NS	NS	NS
For comparison of varieties					
CD(P=0.05)	0.67*		26.54	NS	1.42
For comparison of date of sowing					
CD(P=0.05)	0.67*		4.84	3.25	2.80

*= significant , NS= Non significant

The effect of cultivars and dates of sowing showed that maximum seed yield (15.11 qha-1) was recorded with JAKI-9218 followed JG-16 (12.89) and JG-130 (11.82) which was statistically non-significant. The highest seed yield (15.27 qha-1) was obtained under 15th November sown crop. Whereas it was 11.82qha-1 and 9.83qha-1 in 15th December and 15th November sown crop, respectively which was significant. Interaction of varieties and dates of sowing showed the maximum yield on cultivar JAKI-9218 (19.16 qha-1) under 30th November sown crop and minimum in JG-16 (8.41 qha-1) under 15th December sown crop. Maximum 100 grain weight (23.30 g) was obtained under the 30th November date of sowing and minimum in 100 grain weight (20.78 g) under 15th December sown crop. Interaction of varieties and dates of sowing showed maximum 100 grain weight in JAKI 9218 (24.89 g) followed by JG 130 (24.48 g) and whereas it was minimum in JG-16 (16.17 g).

4.6: *In-vitro* evaluation of botanicals and chemicals against *Rhizoctonia bataticola*.

In the present investigation, the efficacy of three botanicals (@20%) and seven fungicides was evaluated against the growth of R. bataticola. Data on mycelial growth was recorded at five and seven days after inoculation.

At five days after inoculation, all tested botanicals significantly inhibited and reduced mycelial colony diameter over the control. Data presented in Table 4.4 showed that significantly maximum mycelium growth inhibition was recorded in *Allium cepa* (100 %) followed by *Z. officinale* (68.65 %) whereas minimum mycelium inhibition was recorded in *D. stramonium* (51.50 %).

Efficacy of 7 fungicides were tested at different concentrations in the laboratory against *R. bataticola* (Table-4.4, Fig.-2). The results obtained indicated that there was significant difference among the fungicides in inhibiting the mycelial growth of *R. bataticola* at 3 days after inoculation. Among the fungicides evaluated, 100 per cent mycelium inhibition was recorded in Carbendazim @ 0.1 %, followed by Propiconazole @ 0.1 %, Mancozeb @ 0.2 %, Hexaconazole @ 0.1 % and Mancozeb + Copper oxychloride @ 0.2 % + 0.2 % whereas it was 97.40 percent with copper oxychloride (0.2%).

Table 4.4: *In-vitro* evaluation of botanicals and fungicides on the growth of *R. bataticola*

Botanicals/ Fungicides	Concentration (Per cent)	Mycelial growth (mm) /days after inoculation	
		5	7
<i>Datura stramonium</i>	20	21.66 (51.50)	42.66 (52.6)
<i>Zinger officinale</i>	20	14 (68.65)	23.6 (73.77)
<i>Allium cepa</i>	20	0 (100)	2.13 (97.63)
Carboxin	0.1	12.6 (71.78)	31.66 (64.8)
Carbendazim	0.1	0 (100)	0 (100)
Propiconazole	0.2	0 (100)	3.26 (96.37)
Hexaconazole	0.1	0 (100)	0 (100)
Mancozeb	0.2	0 (100)	0 (100)
Copper oxychloride	0.2	1.16 (97.40)	5.6 (93.7)
Mancozeb + Copper oxychloride	0.2+0.2	0 (100)	0 (100)
Control		44.66	90
CD at 5%		1.604	2.68
Sem+		0.504	0.903

Data recorded on mycelium inhibition of *R. bataticola* at seven days after inoculation showed that all the screened botanicals significantly inhibited and reduced mycelial colony diameter over the control. Significantly maximum mycelium

growth inhibition was recorded in *A. cepa* (97.63 %) followed by *Z. officinale* (73.77 %), whereas minimum mycelium inhibition was recorded in *D. stramonium* (52.6 %).

After seven days of inoculation, the complete inhibition of radial growth of *R. bataticola* was found in Carbendazim (0.1%), followed by Mancozeb (0.2%), Hexaconazole (0.1%) and Mancozeb + Copper oxychloride (0.2 % + 0.2 %) whereas it was 96.437% and 93.72% with propiconazole (0.1%) and copper oxychloride (0.2%), respectively. The minimum inhibition was found in Carboxin (0.1 %). In the present investigation, tested the efficiency of different botanicals and fungicides was evaluated on the growth of *R. bataticola*. Data presented in Table 4.4 revealed that all the botanicals significantly inhibited the growth *R. bataticola*.

After inoculation all tested botanicals and fungicides significantly reduced the fungal colony diameter whereas one botanical (*Allium cepa*) and five fungicides (Carbenazim, Propiconazole, Mancozeb, Hexaconazole, Mancozeb + Copper Oxychloride) completely inhibited the growth. Maximum growth was found in control (44.66)

At seven day after inoculation out of eight treatments, four fungicides (Carbendazim, Mancozeb, Hexaconazole, Mancozeb + Copper Oxychloride) were absolutely inhibited the fungal growth. In remaining six treatment minimum fungal growth was present in *A. cepa* (2.13mm) followed by Propiconazole (3.26mm), Copper Oxychloride (5.66mm), *Z. officinales* (23.66), Carboxin (31.66mm) and *Datura stramonium* (42.66mm) while maximum growth was recorded in control (90.00mm).

The four fungicides namely (Carbendazim, Mancozeb, Hexaconazole, and Mancozeb+ Copper oxychloride) were statistically at par with *A. cepa* and significantly superior over the other treatment. *Allium cepa* is significantly superior over the *Z. Officinales*, Caboxin and dextrose but statistically at par with Propiconazole and Copper Oxychloride.



Influence of varieties and date of sowing on occurrence of dry root rot and yield and attributing parameters in chickpea (seedling stage)



Influence of varieties and date of sowing on occurrence of dry root rot yield and attributing parameters in chickpea (podding stage)



**Screening of chickpea cultivars in root rot sick
microplots(20 day old crop)**



**Screening of chickpea cultivars in root rot sick microplots
(at podding stage)**



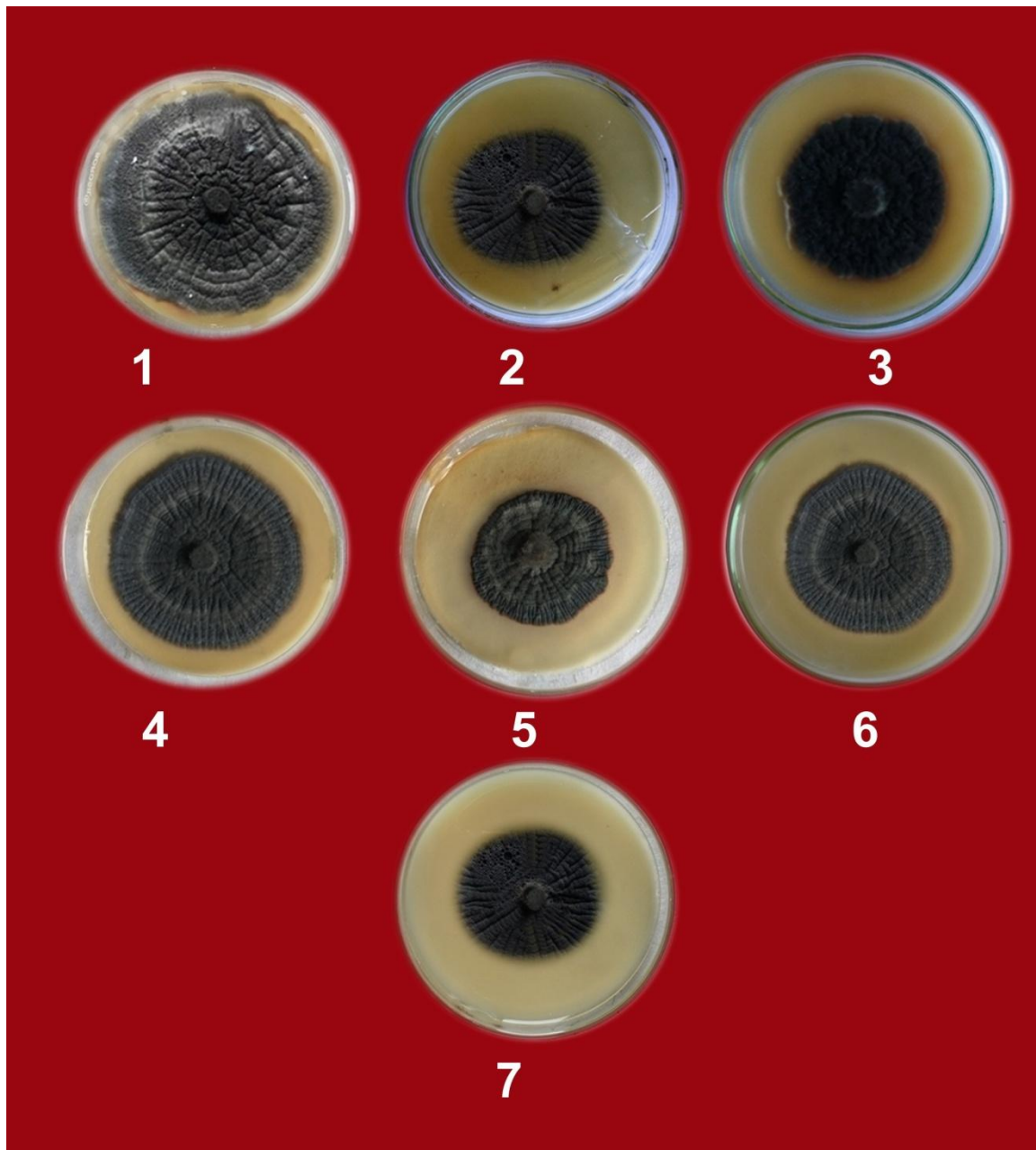
Symptom of dry root rot of chickpea



Rotting of root system

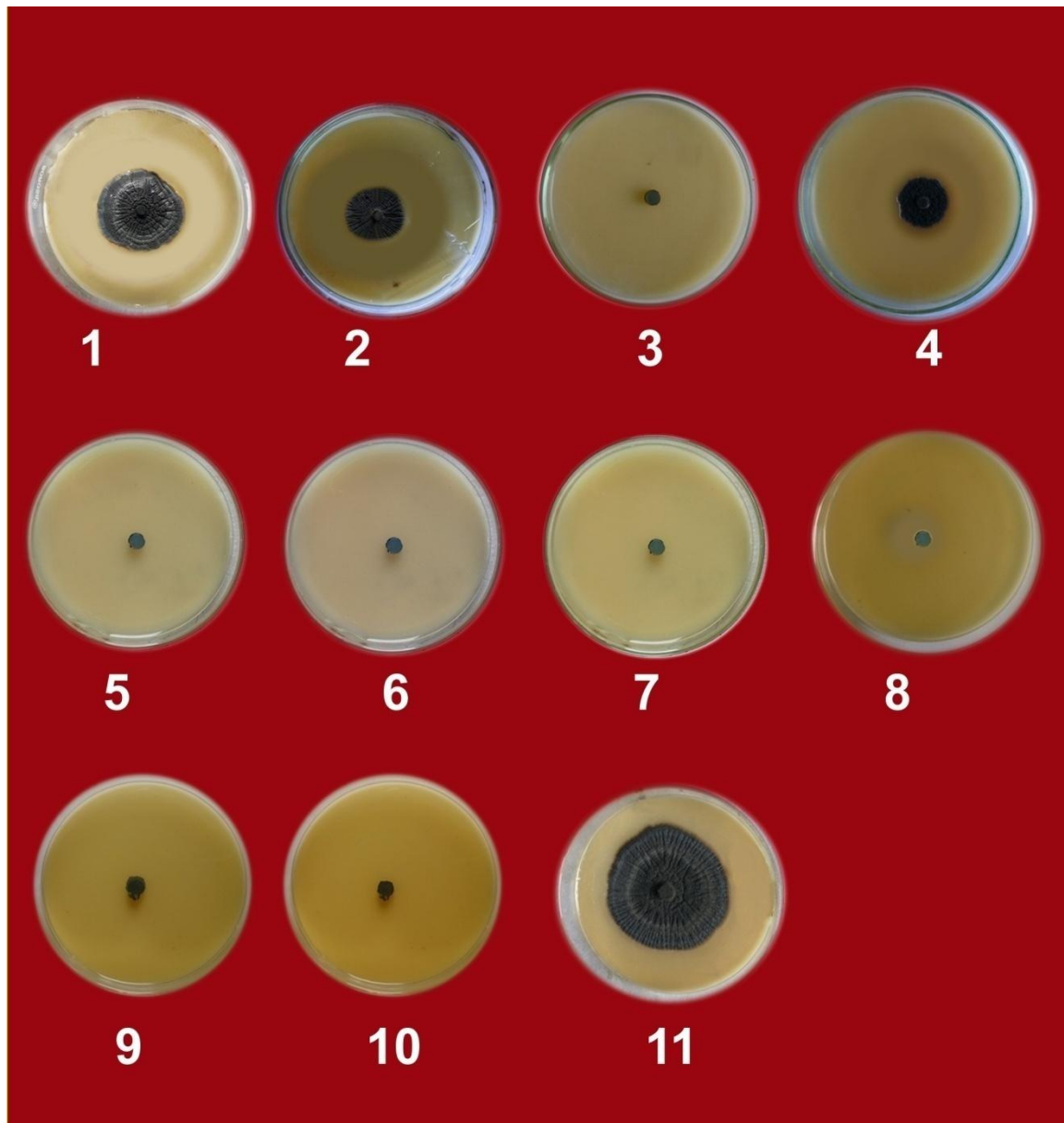


Plate - 1: Mycelial growth of *R. bataticola* on potato dextrose broth
➤ Chickpea seed dipped in *R. bataticola* culture solution



- 1- Potato dextrose agar
- 2- Water agar
- 3- Czapek's Dox agar media
- 4- Chickpea leaf potato dextrose agar
- 5- Sorghum grain agar
- 6- Chickpea potato dextrose agar
- 7- Chickpea grain agar dextrose

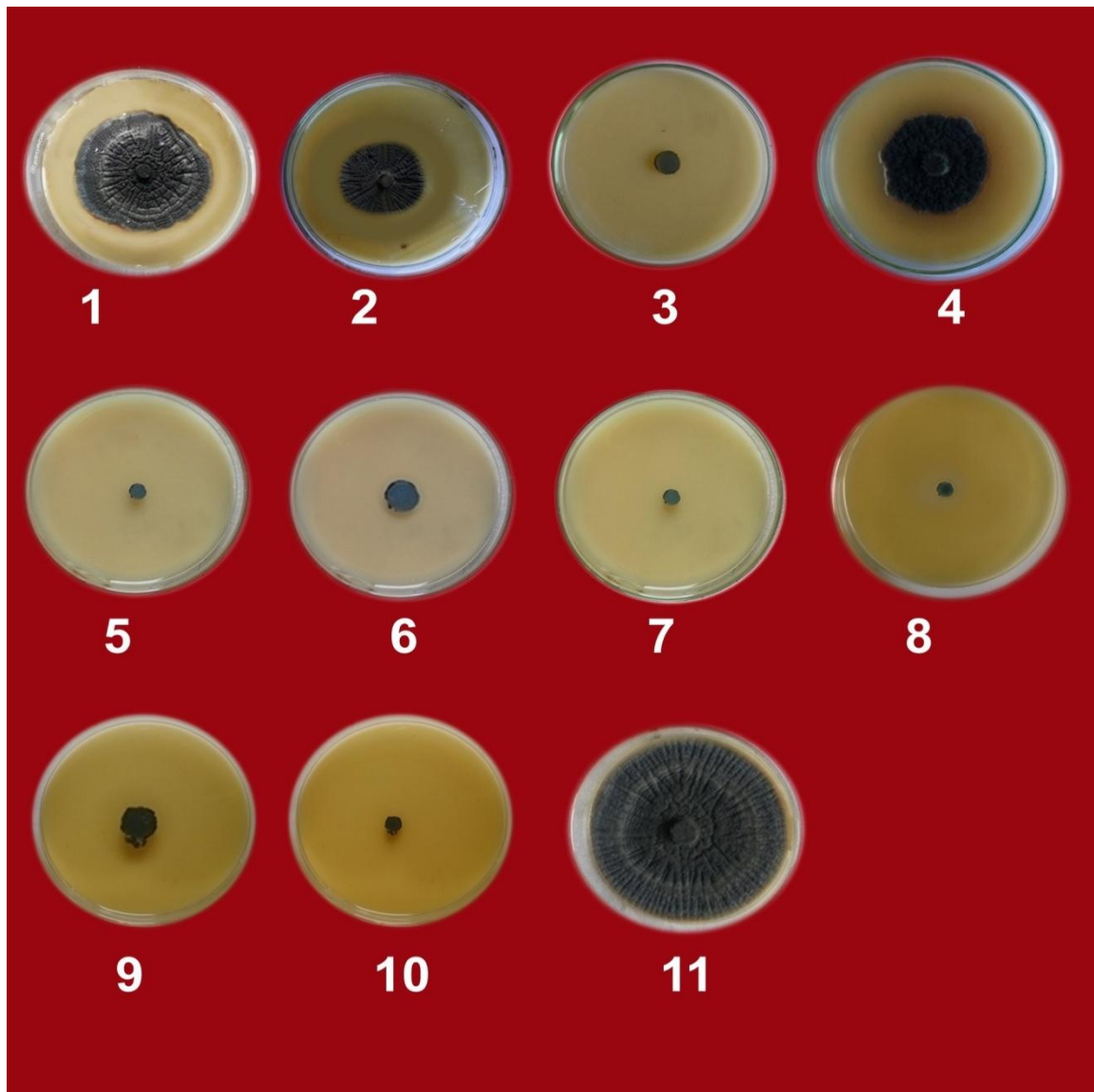
Plate – 3 Evaluation of culture media against growth of *R. bataticola*



- 1- *Datura stromonium*
- 2- *Zingiber officinale*
- 3- *Allium cepa*
- 4- Carboxin
- 5- Carbendazim
- 6- Propiconazole

- 7- Mancozeb
- 8- Hexaconazole
- 9- Copper oxychloride
- 10- Mancozeb + COC
- 11- Control

Plate – 6.1 *In-vitro* evaluation of botanicals and chemicals against growth of *R. bataticola* (5 DAI)



- 1- *Datura stromonium*
- 2- *Zingiber officinale*
- 3- *Allium cepa*
- 4- Carboxin
- 5- Carbendazim
- 6- Propiconazole

- 7- Mancozeb
- 8- Hexaconazole
- 9- Copper oxychloride
- 10- Mancozeb + COC
- 11- Control

Plate – 6.2 *In-vitro* evaluation of botanicals and chemicals against growth of *R. bataticola* (7 DAI)

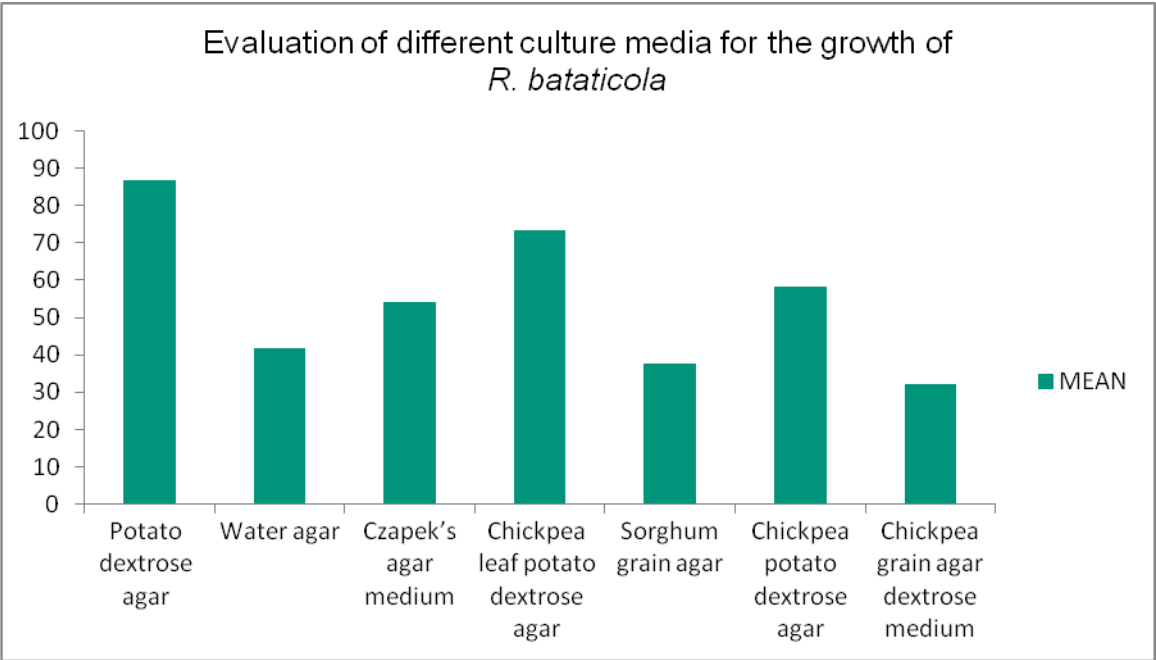


Fig-1. Evaluation of different culture media for the growth of *R. bataticola*

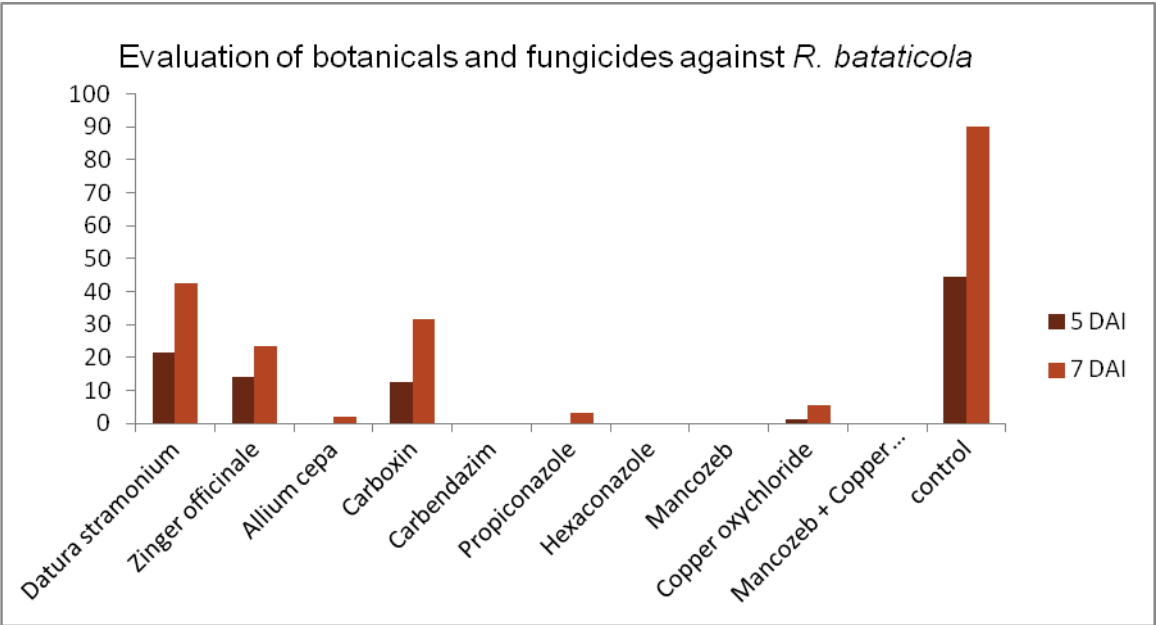


Fig-2. Evaluation of botanicals and fungicides against *R. bataticola*

Chapter - V

DISCUSSION

Chickpea (*Cicer arietinum*) is a legume crop of the family *Fabaceae*, subfamily *Faboideae*. Chickpea, is a self-pollinating diploid ($2n=2x=16$) crop. Chickpea is a Rabi pulse crop and is grown in several countries worldwide as a food source.

Symptoms of dry root rot are most commonly observed in chickpea during post-flowering stage. The leaves and stems of affected plants are usually straw colored, but in some cases the lower leaves and stems are brown. If the plant were pulled out from the soil and examined the basal stem and main root system of diseased plant showed extensive rooting with most of the lateral roots destroyed. The tissues were weakened and break off easily. Dark minute sclerotial bodies on the causal organism can be seen on the roots exposed and also inner side of the bark.

The symptoms resembled with the symptoms as described by Chattopadhyaya and Bathacharya (1967)

The infected plants appear completely dried and is an important component of disease complex that cause root rot and seedling blight in many grain legumes when they are weakened by other stress factor in the absence of host crop. The pathogen survives in soil as a competitive saparophyte on available dead organic matter (Hwang *et al.* 2003)

Seven culture media *viz*; Potato dextrose agar, Czapek's Dox agar medium, Water agar medium, Chickpea leaf potato dextrose agar medium, Sorghum grain agar medium, Chickpea potato dextrose agar medium and Chickpea grain agar dextrose medium, were tested against of *R. bataticola*. Significantly maximum mycelial growth was observed in Potato dextrose agar followed by Chickpea leaf potato dextrose agar, Chickpea potato dextrose agar which were found at par with Czapek's Dox agar medium. Minimum mycelium growth was observed in Chickpea grain agar medium followed by Sorghum grain agar medium, and Water agar medium.

Present findings support the findings of Sharma *et al.* (2013) and Salunkhe *et al.*, (2009). They also reported maximum growth of *R. solani* in Potato dextrose agar followed by Czapek's Dox agar, Corn meal agar, Richard's agar and Asthana and Hawkers medium. *Rhizoctonia bataticola* grew best on Cotton root extract agar, PDA, Richard's agar and Sabouraud's agar (Kulkarni *et al.* 1992)

The effect of eight different solid media was studied on the growth of the pathogen and it was found that potato dextrose agar medium was the best medium for growth of the pathogen followed by standard nutrient agar medium, chickpea root extract agar medium and Colter root extract agar medium whereas the remaining media viz., Oat meal agar, Czapek's Dox agar, Richards agar and Asthana and Hawker's agar supported poor to very poor growth. More or less similar results were also obtained in liquid media tested. Earlier, no such systemic effect was made to find out the best suitable medium for the growth of *R. bataticola*. Very meager information is available on the effect of media on *R. bataticola*. (Post, 1933; and Knox - Davies, (1965)

Zhou *et al.* (2002) evaluated 12 culture media to know the mycelial growth and sclerotial number of *R. solani*. Results showed that cooked Rice straw solution agar, Water agar, PDA and Maltose agar were better media for *R. solani* while Beef extract Peptone agar was not good medium. Richard's agar was the best medium for sclerotial formation, as well as number of sclerotia formed.

The production of chickpea is largely constrained by Fusarium wilt (*Fusarium oxysporum f. sp. Ciceris*); however, recent report indicated that dry root rot DDR is emerging as a potential threat to chickpea production (Pande *et al.* 2004; Sharma *et al.* 2010; Ghose *et al.* 2013). The DDR is caused by *R. bataticola* (taub.) Butler. and is an important component of the disease complex that cause root rot and seedling blight in many grain legumes when they are weakened by other stress factor (Hwang *et al.* 2003)

A total of 261 entries of Chickpea were evaluated under artificial condition against dry root rot of chickpea. It is clear from the data that 4 entries were Resistant, 3 Moderately resistant, 50 Moderately susceptible, 123 Susceptible and remaining 81 were Highly susceptible.

Chickpea genotypes have been screened and identified as resistant to dry root rot by various workers on the basis of *in- vitro* and field conditions (Nene et al., 1981; Reddy et al., 1990; Gupta, 1997; Khalid and Ilyas, 2000 and Gupta et al., 2003). Hundred per cent mortality was observed in the susceptible check (L-550) indicating the high and uniform sickness in the soil. Similar findings have also been reported by Jayalakshami *et al.*, 2009.

Iftikhar and Ilyas (2000) screened 108 chickpea germplasm cultivars against DRR and found only one chickpea line (ICCV 97112) as resistant.

Several germplasm and breeding line were evaluated using Paper towel screening technique and Sick plot screening technique by various researchers.(Nene *et al.* 1981; Gurha *et al.* . 2003;Ashraf *et al.* 2005; Pandey *et al.* 2012).

The incidence of dry root rot of chickpea caused by *R. bataticola* was observed in late during month of October to mid November and the intensity of the disease was high in the month of February and March during late flowering and podding stage.

In the date of sowing experiment three cultivar of chickpea were sown in three dates. In the present findings maximum disease incidence was recorded in JG-16 on 15December sown crop and minimum in JG-130 on 15 November sown crop. disease incidence was minimum in 15th November sown crop followed by JAKI-9218 and JG-16..

On 30th December sown crop minimum disease incidence was recorded in JAKI-9218 followed by JG -130 and JG-16. These data clearly indicate that in early sowing the root rot disease of chickpea was minimum.

Sharma *et al.* (2013) reported that erratic rainfalls and rise in temperature have become more frequent under the changing scenario of climate particularly in semi arid tropics. Dry root rot (DRR) caused by *R. bataticola* was found as a potentially emerging constraint to chickpea production as compared to wilt. Increasing incidence of DRR indicate strong influence of climate change variables such as temperature and moisture on the development of disease. The DRR incidence was significantly affected by high temperature and soil moisture deficit.

Out of four temperature regimes (15^o C, 20^o C, 25^o C and 30^o C) and four moisture levels (40, 60, 80 and 100%), a combination of high temperature (30^oC) and soil moisture content (60%) predisposes chickpea to DRR. The study clearly demonstrates that high temperature coupled with soil moisture deficit is the climate change variable predisposing chickpea to *R. bataticola* infection, colonization and development.

Khan *et al.* (2013) reported that the variation in date of sowing was tested as an effective and economic strategy against dry root rot caused by (*R. bataticola*) and wilt (*Fusarium oxysporum f.sp. ciceris*) disease of chickpea.

The major constraints responsible for this untapped yield potential are in appropriate production practices, viz., usage of low yielding and non – responsive genotypes, pest and disease problems, lack of stress –resistant high yielding genotypes, lack of improved soil and crop management practices and lack of appropriate institutional support (Ramakrishna *et al.*, 2005)

In present investigation three selected botanicals (@ 20%) viz., *Datura stramonium*, *Zingiber officinale*, *Allium cepa* and seven fungicides, viz., Carboxin @0.1%; Mancozeb @0.2%; Copper Oxychloride @ 0.2%; Carbendazim @0.1%; Propiconazole @ 0.1%; Hexaconazole @0.1%; and Mancozeb+ Copper oxychloride @0.2%+0.2% were evaluated against *Rhizoctonia bataticola* under *in-vitro* condition. These four fungicides (viz. Carbendazim, Hexaconazole, and Mancozeb + Copper oxychloride Mancozeb,) were statistically at par with each other in inhibiting the growth of *R. bataticola*. Extract of *Allium cepa* was significantly superior over the other treatments. The present findings are confirmed by Sangeetha and Jahagirdar, (2013).

The phytoextracts of nine plant species viz., garlic, onion, lantana, tulsi, jatropa, arduisi, neem, turmeric and ginger were evaluated *in vitro* by poisoned food technique against *M. phaseolina* causing root rot of sesame. (Savaliya *et al.*, 2015) The extract of garlic cloves (*Allium sativum* L.) was proved excellent with maximum inhibiting mycelial growth (77.65%) and scanty sclerotial formation followed by onion bulb extract (*Allium cepa* L.) (63.98 %) While least growth (32.34%) inhibition was recorded in ginger rhizome extracts

Botanical extracts are biodegradable (Devlin and Zettel, 1999) and their use in crop protection is a practical sustainable alternative. It reduces environmental contamination and health hazards (Grange and Ahmed, 1988). Botanical fungicides are unique because they can be produced easily by the farmers and small industries (Roy *et al.*, 2005).

Prashanthi *et al.*, (2000) reported that Carbendazim (Bavistin) and propiconazole were highly effective with the mean mycelial growth inhibition of 100 and 93.7%, respectively. Carbendazim showed 100% inhibition at all concentrations. Among the non-systemic fungicides, Chlorothalonil SC at 3000 ppm was the most effective (82.45% inhibition of mycelial growth).

Sharma and Gupta (2004) observed that *Carbendazim* treated seed showed the minimum disease incidence (17 to 21 %), of dry root rot of chickpea. The extract of garlic clove (*Allium sativum* L.) was proved excellent with maximum mycelial growth.(Dhingani *et al.*,2013).

Lambhate *et al.*, (2002) tested the efficacy of fungicides against *M. Phaseolina*, root rot pathogen of cotton *in vitro* and reported that Bavistin,Ridomil M Z- 72 and Topsin-M at 0.1, 0.2 and 0.3% showed cent per cent inhibition of mycelia growth of fungus. Khalikar *et al* (2011) reported that Carbendazim 50% WP @ 2g/kg, was superior in reducing pre- and post- emergence seedling rot and root rot disease (*M. Phaseolina*) of Cluster bean.

The general mode of action of systemic fungicides is associated with interference with the electron transport chain influencing the energy budget of the cell, reduction in the biosynthesis of new cell material required for growth and development of the organism and disruption of cell structure and permeability of cell membrane. (Ravichandran and Hegde,2017).

Chapter -VI

SUMMARY, CONCLUSION AND SUGGESTIONS FOR FURTHER WORK

Chickpea (*cicer arietinum* L.) is one of the major grain legumes grown worldwide. It is the rich source of protein (20 to 25%) and also enriches soil fertility by biological nitrogen fixation. It is of prime importance in the Mediterranean basin and South Asia. The crop is vulnerable to a number of air-borne and soil-borne diseases, out of them dry root rot (*Rhizoctonia bataticola*) is emerging in dry and warm regions of India.

Out of seven culture media evaluated against growth of *R. bataticola*. Maximum mycelial growth was observed in Potato dextrose agar followed by Chickpea leaf potato dextrose agar, Chickpea potato dextrose agar which were found at par with Czapek's agar medium. Minimum mycelium growth was observed in Chickpea grain agar medium.

A total of 261 entries of chickpea were evaluated under natural field condition to find out their reaction against dry root rot of chickpea. Four entries were Resistant, 3 Moderately resistant, 50 Moderately susceptible, 123 Susceptible and at last 81 entries were Highly susceptible.

Effect of cultivars and date of sowing on the basis of disease incidence of dry root rot was quite evident. The maximum incidence of *R. bataticola* was recorded in JG-16 and minimum on JAKI 9218

The maximum disease incidence was observed in 15th December sowing and whereas it was minimum disease in 15 November sowing. In the interaction of varieties/cultivars and dates of sowing, cultivars JG-16 showed the maximum disease incidence in 15th December sowing and minimum in JG-130 under 15th November sown crop.

The effect of cultivars and dates of sowing affected the number of pods/plant. The maximum pods were observed with cultivar JAKI-9218, and minimum with JG-16. Maximum pods /plant were observed on 30th November, 15th December and minimum in 15th November sown crop. Interaction of varieties and dates of sowing revealed that maximum pods/plant on cultivar JAKI-9218 under 30th November sown crop and minimum in JG-130 under 15th December sown crop.

Alium cepa and five fungicides (Carbenazim, Prepiconazole, Mencozeb, Hexaconazole, Mencozeb + Copper Oxychloride) completely inhibited the growth of *R. bataticola*

Conclusion:

Potato dextrose agar was the best medium for the growth of *R bataticola*.

Out of 261 entries of chickpea evaluated, 4 entries were resistant , 3 moderately resistant, 50 moderately susceptible, 123 susceptible, and remaining 81 highly susceptible.

Maximum disease incidence was recorded in JG-16 on 15December sown crop and minimum in JG-130 on 15 November sowing, On 15 November sowing minimum disease incidence was with JG-130. On 30December sowing minimum disease incidence was recorded in JAKI-9218.

Alium cepa and five fungicide (Carbenazim, Prepiconazole, Mencozeb, Hexaconazole, Mencozeb + Copper Oxychloride) completely inhibited the growth of *R. bataticola*.

6.3: SUGGESTIONS FOR FURTHER WORK.

- Morphological, physiological and biochemical factors responsible for disease resistance needs investigation.
- More work needed with respect to knowledge of infection process and host defence mechanisms will help in devising effective strategies to DRR.
- There is also a need to map the genetic profile of the few chickpea accession that have shown traces of resistance against DRR.
- Study the defence mechanism of *R.bataticola* in chickpea for DRR.

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APPENDICES**1-Evaluation of different culture media**

Source of Variation	DF	Sum of Squares	Mean Squares	F-Calculated	Significance
Treatment	6	4,939.33	823.222	3.699	0.02049
Error	14	3,115.33	222.524		
Total	20	8,054.67			

2- Date of Sowing

ANOVA								
S.V	Df	SS	MSS	Fcal	Ftab	SIG/NS	S.Em.	CD
REP	2	1.19	0.60	1.32	3.63	NS		
Date Of Sow. (D)	2	312.70	156.35	345.88	3.63	SIG	0.22	0.67
Variety (V)	2	238.99	119.49	264.35	3.63	SIG	0.22	0.67
Interaction (D*V)	4	4.26	1.06	2.36	3.01	NS	0.39	1.16
Error	16	7.23	0.45					
Total	26	564.38						

3-Evaluation of botanicals and fungicides**At 5 DAI**

Source of Variation	DF	Sum of Squares	Mean Squares	F-Calculated	Significance
Treatment	10	6,048.71	604.871	732.505	0
Error	22	18.167	0.826		
Total	32	6,066.88			

7 DAI

Source of Variation	DF	Sum of Squares	Mean Squares	F-Calculated	Significance
Treatment	10	23,784.49	2,378.45	1,070.06	0
Error	22	48.9	2.223		
Total	32	23,833.39			

VITA

The author of this thesis Rekha Shrivastav D/o Sh. Laljeet Shrivastav was born on 14 June, 1994 at District- Shivpuri, Madhaya Pradesh.

She completed Primary education at his native place, High School education with second division in the 2009 from Pohari Public School, District- Shivpuri, Madhaya Pradesh and Intermediate education in the year 2011 with first division from S.N. School, District- Shivpuri, Madhaya Pradesh and completed B.Sc. (Ag.) Degree in 2016 with 7.16 O.G.P.A. from RVSKVV Gwalior (M.P.).

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During all the period of education, from schooling to post graduation she was very sincere and honest towards studies and works.

(REKHA SHRIVASTAV)